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Finding the range of beta decay using the changes in radiation magnitude from shielding and distance

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The distance dependence of beta decay, a type of ionizing radiation, is measured using a Geiger counter. The decline in radiation magnitude as distance increases is measured to be a $1/r^2$ relationship. During the experiment, a plastic sheet is used to determine the extent to which plastic shielding affects radiation magnitude. The experiment concludes that plastic shielding has a large effect on the measured magnitude, but would not be an ideal material for shielding.

I. INTRODUCTION

Even with only a handful of mishaps, nuclear physics has become synonymous with events of disaster even with its large potential for many applications. For this reason, it is important to understand radiation and the relationships surrounding it.

Beta decay, a type of ionizing radiation, was discovered by Sir Ernest B. Rutherford in 1899 [1]. Dr. Hans Geiger, a scientist working under Rutherford, developed the Geiger counter, an instrument that originally measured the magnitude of alpha radiation by counting each reaction [2]. In 1925, Geiger updated the original design in collaboration with one of his graduate students, Walter Müller, to be more efficient and detect many types of ionizing radiation, including beta decay.

To better understand beta decay, an experiment is conducted to analyze the dependence of radiation dosages on distance from the source. Multiple radioactive samples are tested at different distances to determine the relationship that defines this distance dependence. Furthermore, to analyze the effect of shielding on radiation dosage, each sample will be tested with and without a plastic sheet.

II. RADIATION-DISTANCE RELATIONSHIP

Beta decay occurs in unstable isotopes of atomic nuclei. There are two types, both of which result in a more stable atom [3]. In beta-plus decay, a proton turns into a neutron, causing the atom to emit a neutrino and a positron. In beta-minus decay, a neutron becomes a proton, so the atom emits an antineutrino and an electron. In both cases, the atoms change elements but maintain the same number of particles.

An experiment is performed to determine the relationship between radiation and distance from a radioactive sample. A Geiger-Müller detector is directed at the given sample and counts each time beta decay occurs in the sample. The radiation count is measured for about 90 s for multiple trials to reduce error. The distance between the Geiger counter and the sample is recorded in each trial and tested across a range of distances. This experiment was performed using a sample of Cesium 137 (Cs-137), which uses beta-minus decay to turn into Barium 137 (Ba-137) [4], and a sample of Strontium 90 (Sr-90), which uses beta-minus decay to become Yttrium 90 (Y-90) [5].

This experiment was also performed to test how effective plastic shielding is in restricting radiation. To do this, some trials were performed with an additional thin sheet of plastic inserted between the radioactive sample and the Geiger counter.

Figure 1 gives the mean count of each trial for multiple distances. The data was fitted with curves based on a $1/r^2$ relationship. Theoretically, since the radiation

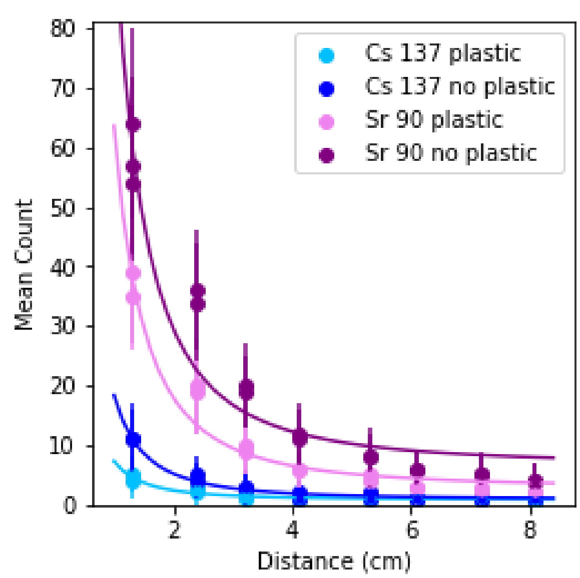

FIG. 1. Mean radiation count over time compared with distance from the Geiger counter.

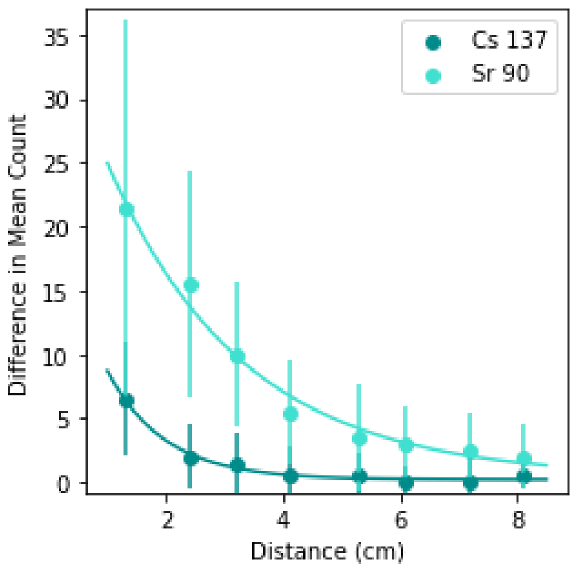

FIG. 2. The difference in average counts between trials with and without plastic shielding.

is emitted in all three dimensions, the radiation magnitude should decrease with the $1/r^2$ relationship. Since the curves generally lie within the range of uncertainty of each measurement, the measured and theoretical relationships match.

Perhaps a more interesting result is whether the plastic was effective in providing shielding from the radiation. From inspecting figure 1, the count decreased significantly when the plastic sheet was used. This relationship is further examined in figure 2.

At the shortest distance measured, the average count for Cs-137 decreased from 11 to 4.5 when plastic was added, an almost 60% drop. At the same distance, the average count for Sr-90 decreased from 58 to 37 with plastic, an almost 40% drop.

The difference in average count decreases as the sample increases in distance from the Geiger counter. For the longest distance tested, the average count of Cs-137 decreased by only half a count, while Sr-90 decreased by 2 counts. In further analyzing this trend, the plastic causes the detected radiation to be significantly lower than the trials without the shielding. However, as the raw amount of detected radiation decreases, so does the raw effect of the shielding. Since the Sr-90 sample is more radioactive, the greater effect the plastic has on detected radiation, even at larger distances, makes sense.

This result is rather significant in the the context of testing different materials for potential use for shielding. From this experiment, it is shown that even a thin layer of plastic makes a large difference in the amount of radiation received. This knowledge helps explain why the radioactive samples used can be stored in a thin plastic case; between the relatively low radioactivity and the minor amounts of shielding, not enough radiation leaves the case to be significant. However, the effect of the plastic sheet was not particularly great if analyzed as a material that specifically needed to provide shielding. For example, if the samples were more radioactive or greater in quantity, a material such as lead would be much preferred, even for such a small case.

III. CONCLUSION

By measuring radiation from multiple distances away from the sources, sufficient data was acquired to determine a mathematical relationship describing the distance dependence of beta decay radiation. Specifically, the data pointed to a $1/r^2$ relationship, where the radiation magnitude decreases by approximately this much as distance from the source increases. Furthermore, sheets of plastic were used in some trials to determine the effect of shielding from plastic. While the plastic provided a radiation magnitude decrease of about half, the plastic was not perfect at providing shielding.

Future work would involve testing the shielding effectiveness of other materials, such as wood and lead. Other possible directions would include testing other samples of radioactive materials to determine if the plastic shielding is just as effective for other strengths of radioactivity. Understanding how well different materials work for shielding and the distance dependence of radioactivity is vital to understanding how to be safe around radioactive materials and the continuation of their study and use.

Appendix A: Uncertainty analysis

The data displayed in figure 1 was recorded by the Pasco computer system as the mean of the number of counts across the time tested. The uncertainty is the recorded standard deviation within each trial. The error bars shown on the plot are the overlapping error bars for individual measurements of repeated trials.

Data analyzed in figure 2 is the difference between the average number of counts of repeated trials with plastic and those without plastic. The uncertainty was calculated by using the error propagation formula for averages:

$$\delta = \sqrt{\frac{1}{4}\left(\sigma_1^2 + \sigma_2^2\right)}. \tag{A1}$$

This gives the average uncertainty between similar trials. Then, when the difference between trials with and without plastic was found, the uncertainty was found with the error propagation formula for addition:

$$\delta = \sqrt{\sigma_{\text{no plastic}}^2 + \sigma_{\text{plastic}}^2}. \tag{A2}$$

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[2] June 1911: Invention of the geiger counter (2012).
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[4] Epa facts about cesium-137.
[5] Radioisotope brief: Strontium-90 (2018).